

ANALOG CMOS CIRCUIT DESIGN SS26 - PRACTICAL EXERCISE II

Operational Amplifier Testbenches

Tutors

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Analog CMOS Circuit Design SS26

Practical Exercise II – Amplifier Testbenches

Group

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No review without this standard form!



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PRACTICAL EXERCISE II

The practical exercise *Analog CMOS Circuit Design* is intended to familiarize you with the design of analog integrated circuits. Based on the example of a single-stage operational amplifier shown in Figure 1, typical characterization tasks will be performed in this exercise. In order to verify the design parameters of a circuit, a thorough understanding of the characterization is a must. This exercise is the first step towards circuit design by performing a DC bias point, AC, and transient simulations of an operational amplifier.

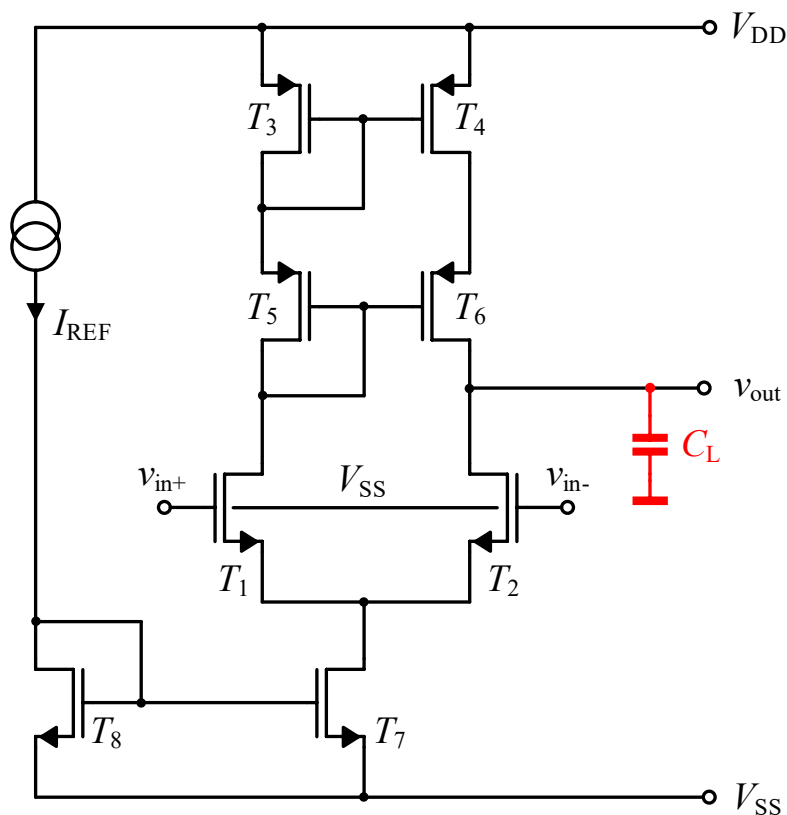


Figure 1: Telescopic Operational Amplifier.

OPAMP Specifications

Load capacitance C_L	10pF
Reference Current I_{ref}	10μA
Nominal supply voltage	$V_{DD} = 3.3V, V_{SS} = 0V$



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PRACTICAL EXERCISE II

T1: DC Bias Point and Offset

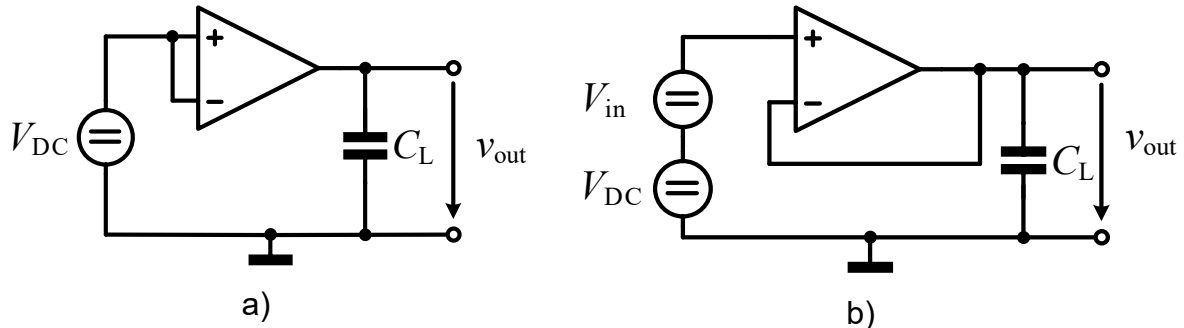


Figure 2: DC bias point testbenches

- Determine output voltage for Implementation a) and b) setting $V_{DC} = V_{DD}/2$.
- Sweep the input range of the amplifier $V_{DC} = 0 \dots V_{DD}$ with $v_{in} = 0$ mV applied and measure the output voltage v_{out} for implementation b). What transistors determine the upper and lower boundary of your input common mode range? State the corresponding equations.
- Use the calculator to implement a function which calculates the DC offset of your amplifier. Determine the offset by setting the v_{in} to 0 and $V_{DC} = V_{DD}/2$. Do you assume the simulated value to be reasonable? What's causing the offset? State the calculator expression.
- Why is implementation b) more suitable to determine the offset?

T2: AC Transfer Function

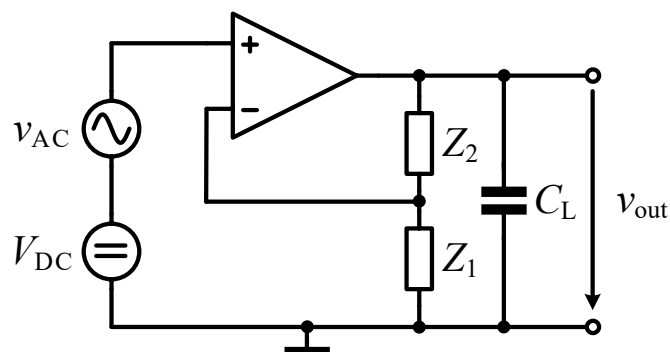


Figure 3: AC testbench



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- a) Derive the AC transfer function $H(s) = \frac{v_{out}}{v_{in}}$ of the OPAMP as a function of the impedances Z_1 and Z_2 (R, L, C) assuming a finite OPAMP gain A_{DC} .
- b) Determine a suitable set of impedances which results in a DC gain of unity and an AC gain equivalent to the open-loop gain of the amplifier. Calculate the transfer function.
- c) How many poles and zeros are realized by your chosen implementation?
- d) Choose appropriate values for both impedances.
- e) Plot the transfer function of the complete amplifier to verify the behavior of your testbench. Indicate the Poles/Zeros corresponding to the amplifier and the external circuitry.
- f) Open the calculator to implement a function which automatically calculates the DC gain, UGBW and Phase Margin from the frequency sweep. State the expression.
- g) Determine the largest load capacitor the amplifier can drive.

T3: Power Supply Rejection

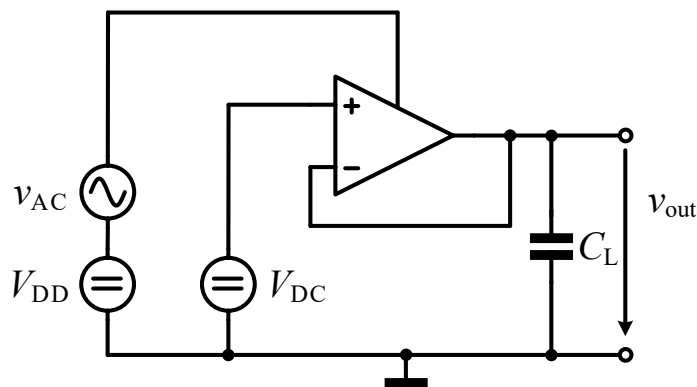


Figure 4: PSR testbench

- a) While keeping the power supply biased at the nominal value, superimpose an AC signal on top and record the output response. What effects dominate for high frequencies?
- b) Extract the DC Power Supply Rejection PSR in dBV/V from your simulation.



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PRACTICAL EXERCISE II

T4: Step Response

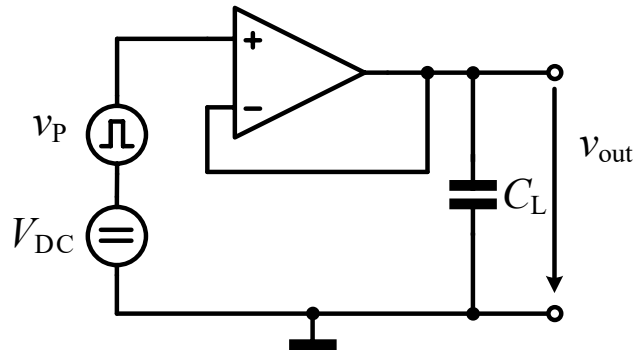


Figure 5: Slew Rate testbench

- a) Apply steps v_p of different sizes (0,20,50,100,200,400,600 mV) to the input, while keeping it biased at $V_{DC} = V_{CM}$. Plot the transient output voltage v_{out} of the amplifier.
- b) Determine the current conducted by the input transistors T1 and T2 during the pulse events (rising & falling edges).
- c) Open the calculator to implement a function which calculates the Slew Rate $SR = \frac{\partial v_{out}}{\partial t}$ in (10-90% steady state output amplitude) V/ μ s for the falling and rising edge of the output signal and extract the corresponding values for the different input step sizes. State the expression.
- d) Why is the slewing behavior of the amplifier related to the magnitude of the input step? Shortly explain your findings!
- e) Estimate the boundary voltage between the two operating regions.

